moving freely in a symmetrical potential well, a situation which is valid only for nuclei near closed shells. The angular momentum of an odd-A deformed nucleus is due both to the rotational angular momentum of the deformed core and to the angular momentum of the odd nucleon.

Consequently the energy levels for such a nucleus are different from those of the symmetric shell model.

This situation was taken into account by S. G. Nilsson, who calculated energy levels for odd nuclei as a function of the nuclear deformation β. Figure 6.8 shows how the energies of the Nilsson levels vary with the deformation β of the potential well. Each shell model level of angular momentum j splits into $j + ½$ levels (called *Nilsson levels* or *states*). Each level may contain up to two nucleons and form the ground state of a rotational band. In addition the undeformed levels ($\beta = 0$) appear in somewhat different order than for the symmetric shell model (Table 6.6). This leads to a reversal in order for some of the levels, e.g. $1f_{5/2}$ and $2p_{3/2}$ (this explains the observed level order for ^{47}Sc, Fig. 6.5). The Nilsson levels are quite different in all characteristics from the shell model states, and their prediction of energies, angular momenta, quantum numbers, and other properties agrees better with experimental data for deformed nuclei than those of any other model.

As an example we may choose $^{23}_{11}$Na, which has a quadrupole moment of 0.101 barn. Assuming the nuclear radius to be $1.1\ A^{1/3} = (a+c)/2$ (fm) we can use (6.38) and (6.39) to calculate a deformation index $\beta = 0.12$. (The value 1.1 for the constant r_o in (4.14) gives better agreement in nuclei where the inertia of the nucleus is involved.) From Figure 6.8 the 11th proton must enter the 3/2 level rather than the $1d_{5/2}$ level as the symmetric shell model indicates in Table 6.6. The experimental spin of 3/2 confirms the Nilsson prediction. Similarly, for the deformed $^{19}_{9}$F and $^{19}_{10}$Ne, we expect from Table 6.8 the odd nucleon to give spin of ½ and not 5/2, as would be obtained from Table 6.6. Again, experiment agrees with the prediction of ½.

6.6. INTERACTION BETWEEN THE NUCLEAR SPIN AND THE ELECTRON STRUCTURE

We have already seen how the spin and orbital angular momentum of the electrons and of the nucleus produce magnetic fields that interact with each other. The field produced by the electrons is much larger than that of the nucleus, and consequently the nuclear spin is oriented in relation to the field produced by the electron shell. By contrast the effects of the nuclear spin on the electron structure is so small that it usually neglected. Nuclear physics has provided us with instruments of such extreme sophistication and resolution that there are many ways of measuring with great accuracy the interaction between the nucleus and the electrons. The result has been research tools of utmost importance, most prominent being the *nuclear magnetic resonance* (nmr) techniques. The

separate disciplines of chemistry, atomic physics, nuclear physics, and solid state physics approach each other closely in such techniques, and an understanding of the theory and experimental methods requires knowledge of all these subjects.

In this section only a few important aspects of the interaction between nuclear spin and electronic structure are reviewed. The methods described are usually not considered to fall within the framework of nuclear chemistry, but in all scientific fields it is important to be able to reach the border and look at developments and techniques used on the other side. Such information is often the seed to further scientific development.

6.6.1. Hyperfine spectra

In §6.3.4 we mentioned that the electrons in the atomic shell have Russell-Saunders coupling, (6.24), $\overrightarrow{J} = \overrightarrow{L} + \overrightarrow{S}$, where $\overrightarrow{J}$ is referred to as the *internal quantum number*. The magnetic field created by the electrons interacts with that caused by the nuclear spin to yield the grand atomic angular momentum vector

$$\overrightarrow{F} = \overrightarrow{J} + \overrightarrow{I} \tag{6.42}$$

The magnitude of this momentum is

$$p_F = \hbar[F(F+1)]^{1/2}, \tag{6.43}$$

but only projections $p_F(z) = 0,\ldots \hbar F$ are permitted on an external (to the atom) field axis. This leads to a large number of possible energy levels, although they are limited by certain selection rules.

The nuclear spin can orient itself in relation to $\overrightarrow{J}$ in

$$2I + 1 \text{ directions if } I \leq J$$

$$2J + 1 \text{ directions if } J \leq I$$

Consider ^{23}Na as an example (Fig. 6.9);[3] it has a nuclear spin $I = 3/2$. The yellow sodium line of 589.6 nm is caused by de-excitation of its electronically excited p state to the ground state $2s_{½}$.

The difference between the two p-states is very small, only 0.0022 eV (or about 0.6 nm), and can only be observed with high resolution (*fine spectra*). To each of these three levels the nuclear spin I has to be added, yielding the quantum number F according to (6.31). It is easy to determine that the level number rule holds, e.g. for $I = J$ we must have $2I + 1 = 4$ possible levels. These levels can be observed in optical spectrometers only at extremely high resolution (*hyperfine spectrum*, hfs). The energy separation between the hfs lines depends on the nuclear magnetic dipole μ_I, the spin value I, and the strength

[3] The letters S and P stand for $\overrightarrow{L} = 0$ ($\Sigma l_j = 0$) and 1, respectively. The superscript 2 refers to the number of possible S-values; thus $2S + 1 = 2$, i.e. $S = \Sigma s_i = 2$. The subscript gives the $\overrightarrow{J}$-value.

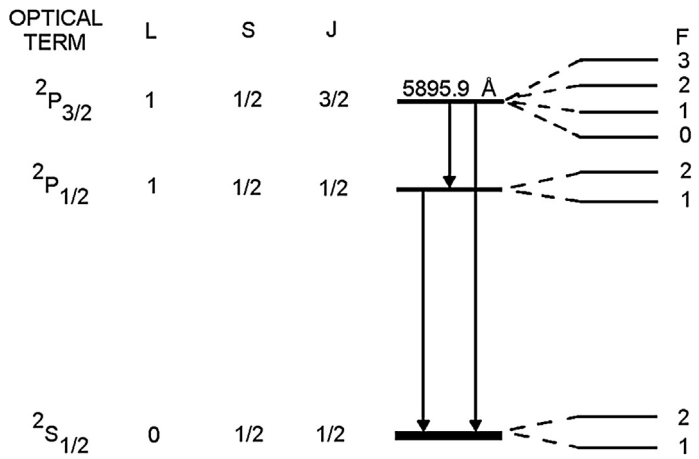


Figure 6.9 The development of hyperfine lines in an optical spectrum of sodium due to the nuclear spin *I*, which enters through $\vec{F} = \vec{J} + \vec{I}$.

of the magnetic field produced at the nucleus, as discussed in §6.2.4 (see also §6.6.3). The energy separation is very small, on the order of 10^{-5} eV, corresponding to a wavelength difference of about 1/1000 of a nm. Even if there is great uncertainty in the energy determination of the levels, simply counting the number of hyperfine lines for a certain electronic $\vec{J}$-level gives the value of the nuclear spin, because $-I \leq m_I \leq I$ (number of m_I values $= 2I + 1$).

At very high resolutions it may be possible to determine μ_I from hfs, and from this to calculate *I*. The hyperfine splitting and shifting of optical lines have yielded important information not only about nuclear spins and magnetic moments, but also about the electric charge distribution and radius of the nucleus.

6.6.2. Atomic beams

The hyperfine spectra are obtained from light sources in the absence of any external (to the atom) magnetic field. If the source is placed between the poles of a magnet, whose strength *B* is progressively increased, the following sequence of changes takes place (see Table 6.2, electron-nucleon interaction).

Suppose *B* is increased slowly to 10^{-2} T. Each hyperfine level *F* is found to split into $2F + 1$ levels, since the quantum mechanical rule of permitted projections on an external field vector comes into operation. These permitted projections can vary from zero to a maximum value of $\hbar F$; for $F = 3$ seven new lines are obtained. Further increases in *B* to 10 T leads to a decoupling of *F* into its components *J* and *I* (Fig. 6.10). For each projection of *J* on the field vector there are $2I + 1$ lines $(-I \ldots 0 \ldots +I)$, giving

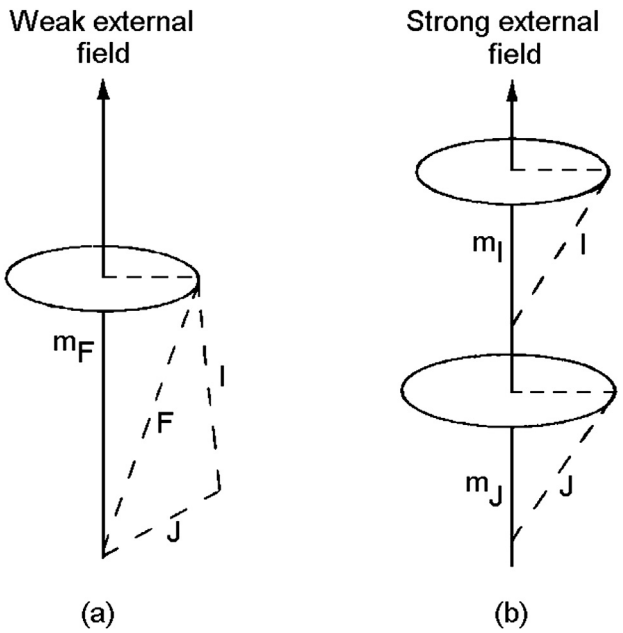


Figure 6.10 (a) Coupling of electronic and nuclear angular moments J and I in a very weak magnetic field. Their sum F forms the quantum number m_F along the direction of the magnetic field, the maximum magnitude of which is $p_F(z) = \hbar F$. In a strong magnetic field the electronic and nuclear angular moments decouple, so that I and J act independently, giving projections m_j and m_i on the field vector.

altogether $(2J + 1)(2I + 1)$ lines. The splitting of the spectral lines in a weak magnetic field is called the *Zeeman effect*.

The decoupling of the angular momentum of the atom into its electronic and nuclear components is used in the *atomic beam apparatus* to determine nuclear magnetic moments and spin values. A beam of atoms, produced in an oven, is allowed to enter a tube along which a series of magnets (usually three) have been placed. The magnetic field splits the atomic beam into several component beams, each containing only atoms which have the same values for all the quantum numbers. Between the magnets there is a small coil connected to a high frequency oscillator, which produces a weak oscillating magnetic field. When this oscillator is tuned to an energy $\mathbf{h}\upsilon$, which exactly matches the energy difference between two quantum states of the atom, energy may be absorbed producing a transition from one of the states to another. Subsequently the atoms are deflected differently by the magnetic field than they were before the energy absorption. By a combination of homogeneous and heterogeneous magnetic fields the atomic beam apparatus allows only those atoms which have absorbed the energy quantum $\mathbf{h}\upsilon$ to reach the detector. From the properties of the magnetic fields and the geometric dimensions and frequency of the instrument, the magnetic moment of the atom in different quantum

states can be determined. This allows calculation of the magnetic moment and spin of the nucleus. This technique is of interest to the nuclear chemist because spin values can be obtained for short-lived nuclei in submicroscopic amounts. In contrast the hfs technique requires macroscopic amounts of atoms.

6.6.3. Nuclear magnetic resonance[4]

When an atom is placed in an external magnetic field (strength B) so that J and I decouple (cf. (6.42) and (6.43)), the nuclear magnetic moment vector μ_I must precess around the field direction, with the components in the direction of the field restricted to

$$\mu_I = g_I \, \mathbf{B}_\mathrm{n} \, m_I \tag{6.44}$$

In the external field the states with different m_I have slightly different energies. The potential magnetic energy of the nucleus is

$$E_\mathrm{magn} = -\vec{\mu}_\mathrm{I} \vec{B} = -g_\mathrm{I} \, \mathbf{B}_\mathrm{n} \, m_I \, B \tag{6.45}$$

The energy spacing between two adjacent levels is:

$$\Delta E = g_I \, \mathbf{B}_\mathrm{n} \, B \tag{6.46}$$

because $\Delta m_I = \pm 1$. For example, consider the case of the nucleus ^{19}F for which $g_I = 5.256$ in a field of 1 T, $\Delta E = 5.256 \times 5.0505 \times 10^{-27} \times 1 = 2.653 \times 10^{-26}$ J. The frequency of electromagnetic radiation corresponding to this energy is 4.0×10^7 s^{-1} or 40 MHz. This lies in the short wave length region, $\lambda = 7.5$ m. This frequency is also the same as that of the Larmor precession, as given by (6.32).

These relations can be used to calculate the nuclear magnetic moment, if I is known, or vice versa. Figure 6.11 shows the results of an experiment in which a sample has been placed in a variable magnetic field containing two coils, one connected to a radio transmitter operating at 5 MHz and the other to an amplifier. The sample is a glass tube (B, O, Na, Al, Si atoms) containing a piece of copper alloy (Cu, Al) in water (H, D, O). By varying the magnetic field a number of resonances are observed. For example, for ^{23}Na one has a resonance at 0.443 T. With (6.46) we can calculate $g_I = 1.48$, and, consequently, the magnetic moment of ^{23}Na is 2.218 (Table 6.3). Further, for ^{23}Na, $m_I = 3/2$ and $I = 3/2$.

The magnetic field experienced by the nucleus is not exactly equal to the external field because of the shielding effect of the electron shell, even if I and J are decoupled. Although this shielding of the nucleus is very small, about 10^{-5} B, it can still easily be detected with modern equipment. The shielding effect depends on the electronic structure.

The structural information that can be provided by this method is very detailed, and a new and deeper insight in chemical bonding and molecular structure is provided. The

[4] Today usually called Magnetic Resonance when used in medicine, etc.

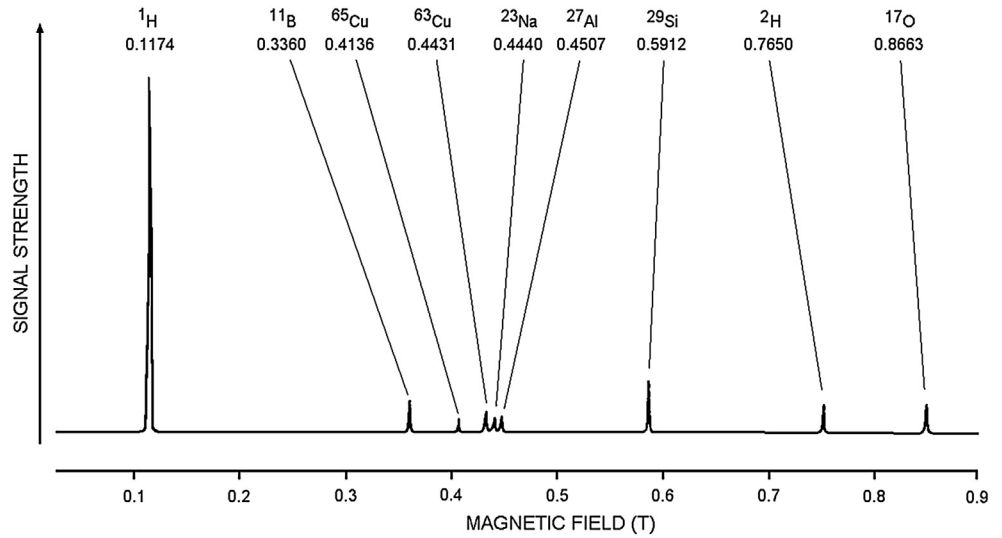


Figure 6.11 Nmr spectrum for a glass vessel containing water and some copper. The magnetic field is given in tesla (T).

nmr technique has therefore become a central tool for the investigation of chemical structures in solids, liquids, and the gaseous state. Tomography by nmr is also a valuable tool for medical imaging. To avoid the unease of the patients caused by the word "nuclear", nmr in hospitals is now usually called magnetic resonance, mr.

6.7. RADIOACTIVE DECAY AND NUCLEAR STRUCTURE

In the preceding section we have described three methods of determining nuclear spin - one optical and two magnetic. The nuclear spin plays a central role in forming the nuclear energy states. It is therefore to be expected that it also should be of importance in nuclear reactions and in radioactive decay. Let us consider some rules for the lifetimes of unstable nuclei, for their permitted modes of decay, and for the role of nuclear spin. Knowing these rules, it is, for example, possible from a decay scheme to predict the spin states of levels which have not been measured.

6.7.1. Gamma-decay

Photons are emitted in the transition of a nucleus from a higher energy state (level) to a lower

$$E_\gamma = \mathbf{h}\upsilon = E_\mathrm{f} - E_\mathrm{i} \tag{6.47}$$

where f and i refer to the final and initial states. Because the photon has spin 1, de-excitation through γ-emission is always accompanied by a spin change $\Delta I = 1$. This